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Serial Ultrasonography for the Assessment of Healing of Lower Extremity Bone Stress Injury and Correlation With Return to Sport/Exercise

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Background: Lower extremity bone stress injuries (BSIs) are common among athletes who participate in high-impact activities. Conventional imaging is limited in assessing healing of BSIs.

Hypothesis: Serial ultrasonography (US) can identify changes in appearance of lower extremity BSIs over time that can be correlated with symptoms and return to exercise/sport.

Study Design: Cohort observational study.

Level of Evidence: Level 3.

Methods: Adults 18 to 50 years old with a recent exercise-associated BSI of distal tibia/fibula or metatarsals diagnosed by magnetic resonance imaging (MRI) were enrolled. US was performed every 2 weeks for 12 weeks. The sonographic appearance (soft tissue edema, periosteal reaction, hyperemia on power Doppler, callus) was correlated with the numerical rating scale (NRS) for pain and ability to return to sport/exercise.

Results: A total of 30 patients were enrolled (mean age, 35.3 ± 7.7 years; 21 [70.0%] female). The tibia was most frequently affected (n = 15, 50.0%), followed by metatarsals (n = 14, 46.7%) and fibula (n = 1, 3.3%). At week 4, 25 of 30 (83.3%) had at least 1 US finding associated with the BSI. The degree of hyperemia was correlated with NRS at weeks 4 and 6 (Spearman correlations [ρ] 0.45 [0.09, 0.69] and 0.42 [0.07, 0.67], respectively), as well as return to sport/exercise at week 6 (ρ -0.45 [-0.68, -0.09]). US soft tissue edema was also correlated with NRS at week 6 (ρ 0.38 [0.02, 0.65]).

Conclusion: Serial US of lower extremity BSIs can provide objective measures of healing. US findings were correlated with clinical outcomes at multiple timepoints.

Clinical Relevance: US may have advantages over conventional imaging for monitoring healing of lower extremity BSIs. Further research is needed to better understand the prognostic value of these sonographic indicators of BSI healing and role in assessing readiness for return to sport/exercise.

Keywords: bone stress injury; magnetic resonance imaging; return to play; ultrasonography

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Bone stress injuries (BSIs) are caused by repetitive submaximal loading of bone that can cause a stress reaction and may progress to a stress fracture.[11,21] Lower extremity BSIs are common among athletes, in particular those whose sport requires repetitive running and jumping.[11,12,16] The most common sites for lower extremity BSI vary somewhat across sports, though the vast majority of BSIs in athletes are of the tibia, fibula, and metatarsals.[4,7,11,20]

Ultrasonography (US) has been studied as an alternative imaging modality for diagnosing BSIs.[14,22] Sonographic findings of BSI include cortical disruption, periosteal elevation with adjacent hypoechoic fluid collection, and hyperemia on power Doppler sonography.[5,10,17] Evaluation of fracture healing has also been demonstrated, although primarily in the context of traumatic fractures.[6,8,15,19] The evidence for US to monitor BSI healing is limited to a case report describing findings of serial sonography of a 68-year-old woman with a metatarsal stress fracture.[2] Therefore, the sonographic appearance of BSIs has unknown value with regard to assessing healing and guiding return to sport/exercise.

The purpose of this study was to evaluate the use of serial US to assess healing of lower extremity BSIs and to determine the relationship between sonographic changes and improvement in clinical symptoms.

METHODS

The study was approved by the institutional review board of Hospital for Special Surgery, New York. This was a prospective observational study of adults with exercise-related BSIs conducted at Hospital for Special Surgery. Enrollment occurred between February 1, 2018 and December 31, 2020. Data collection was completed in February 2021. Inclusion criteria were age 18 to 50 years old; exercise-associated BSI in the metatarsals, distal tibia, or distal fibula (involving the distal half of the bone) diagnosed by magnetic resonance imaging (MRI); and enrolled within 4 weeks of completing the MRI scan. Exclusion criteria were >1 BSI at time of enrollment, BSI associated with hardware from previous surgery, osteopenia or osteoporosis, and hyperparathyroidism or other metabolic bone diseases. Eligibility was verified by self-report and chart review. Subjects were recruited through the clinical practices of the Primary Sports Medicine Service at the Hospital for Special Surgery, local physical therapy practices, and local running clubs.

Data collected included symptom surveys, results of serial US examinations, and results of MRI scans. At the time of enrollment, an initial survey was administered that included demographics (age, sex, race, ethnicity), duration of symptoms, exercise or sport associated with the BSI, numerical rating scale (NRS) for average and worst pain in the past week, history of BSI, and dietary restrictions (Online Appendix 1 - Baseline Survey). Follow-up patient surveys that included NRS for average and worst pain in the past week (Online Appendix 2 - Biweekly Survey) were administered every 2 weeks during the study period. Subjects were asked at each of the follow-up surveys to rate their ability to return to the sport/exercise that caused the initial injury using a 5-point Likert scale.

Sonography was performed by 1 of 4 experienced musculoskeletal sonographers, who were trained in the study protocol by the musculoskeletal ultrasound radiologist coinvestigators. Scans were performed using either a GE Logiq e9 with a 10 to 15 MHz linear transducer (GE HealthCare Technologies Inc) or a Samsung RS 85 with a 12 to 18 MHz linear transducer (Samsung Medison Co). The radiologists assisted the sonographers for performing the scans for the initial subjects enrolled in the study. Depth and scanning frequency were adjusted for each patient and kept uniform on the sequential scans. Power Doppler gain was adjusted to eliminate background noise. Scans were performed every 2 weeks for 12 weeks after the date of the initial MRI scan. For subjects who were enrolled within 2 weeks of the initial MRI scan, a total of 6 US studies were performed (2, 4, 6, 8, 10, and 12 weeks). When enrollment occurred between 2 and 4 weeks after the initial MRI scan, a total of 5 US studies were performed (4, 6, 8, 10, and 12 weeks). The images were reviewed independently by 2 experienced musculoskeletal ultrasound radiologists, blinded to the clinical data. The readers assessed the images for soft tissue edema overlying the BSI, periosteal reaction, and hyperemia on power Doppler sonography (rated as absent, 0; mild, 1; moderate, 2; marked, 3), as well as callus (absent or present). Periosteal reaction was visualized as elevation of the linear echogenic periosteum from the cortical surface, while soft tissue edema was seen as hypoechoic effacement of the fascial planes. Examples of soft tissue edema, periosteal reaction, and hyperemia are found in Online Appendix 3 - Sonographic Images. For discrepancies in the interpretation of the US images within 1 rating, eg, mild (1) and moderate (2), the higher classification was used. For the minority of discrepancies that were a difference of ≥2, eg, mild (1) and marked (3), the images were re-reviewed by both radiologists and re-scored.

MRI was performed before enrollment and repeated at 12 weeks. MRI scans of the distal tibia and fibula included sagittal, axial, and coronal proton density (PD) sequences and axial fat-suppressed T2 images. MRI scans of the metatarsals included short axis, long axis, sagittal PD sequences, and sagittal fat-suppressed T2 images. The MRI scans were assessed by the same 2 musculoskeletal radiologists for marrow edema, soft tissue edema, and periosteal reaction (absent, 0; mild, 1; moderate, 2; marked, 3), as well as cortical fracture line (absent, 0; present, 1). Figure 1 illustrates the US and MRI scan appearance of BSIs by comparing soft tissue edema and periosteal reaction with these 2 modalities.

During the study period, subjects received standard-of-care treatment from the physician who diagnosed the BSI. This included restrictions from running, jumping, and/or other weight-bearing exercises. If there was pain with ambulation, supportive devices, such as crutches, a walking boot, a hard-soled shoe, or a pneumatic splint, were used. Return to sport/exercise was guided by the treating physician.

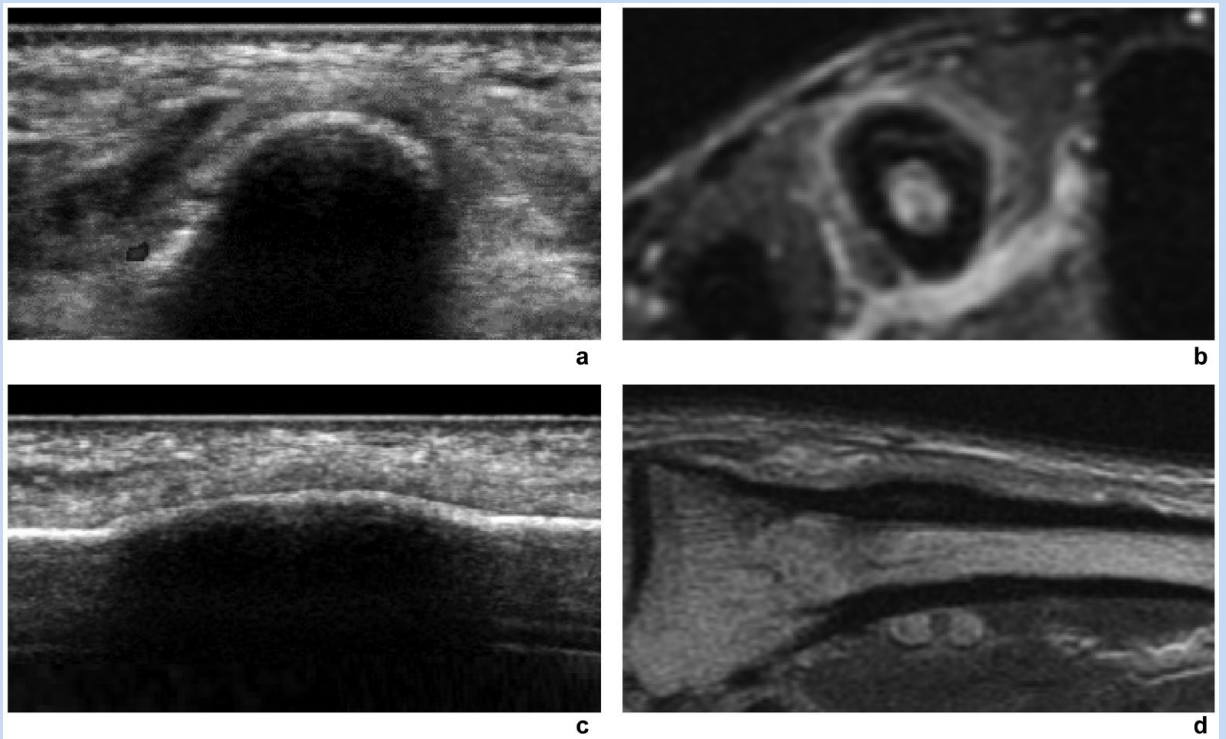

Figure 1. US and MRI comparison for soft tissue edema and periosteal reaction. Subject was a 36-year-old female athlete with a second metatarsal stress fracture. (a) US axial: soft tissue edema (mild). (b) MRI axial fat-suppressed T2: soft tissue edema (mild). (c) US longitudinal: periosteal reaction (moderate). (d) MRI longitudinal proton density: periosteal reaction (moderate). MRI, magnetic resonance imaging; US, ultrasonography.

Statistical Analysis

Descriptive statistics were used for demographics, injury characteristics, and baseline symptom scores with proportions, means and standard deviations for age and NRS, and median and interquartile ranges for the duration of symptoms in weeks. One-factor repeated measures analysis of variance (ANOVA) was used to assess change in NRS for average pain and return to sport/exercise. MRI and sonographic findings were reported as means and standard deviations (or proportions as appropriate) at each time point. Changes in MRI and sonographic findings were assessed using paired sample *t* test and ANOVA, respectively. The last observation carried forward was used to account for missing sonographic data in the ANOVA tests. McNemar's Chi-square test was used to assess the change in cortical fracture identified by MRI. Spearman correlations were used to evaluate the relationship between imaging findings and NRS for average pain and return to sport/exercise at each time point. Microsoft Excel was used for descriptive statistics. JASP (Version 0.14.1) was used for *t* test and ANOVA.

RESULTS

A total of 31 subjects were enrolled. Enrollment was suspended from March through August 2022 due to the COVID-19 pandemic, but no subjects were enrolled at the time. One subject was lost to follow-up after week 2 and was excluded. Table 1 shows the baseline characteristics of the 30 subjects included in the analysis. The sport or exercise associated with the injury was primarily running (n = 27, 90.0%) with the remaining 3 resulting from aerobics, kickboxing, and trampoline. The following medical conditions that can be associated with low bone mineral density were reported: lactose intolerance (n = 4, 13.3%), eating disorder (n = 3, 10.0%), tobacco use (n = 3, 10.0%), and kidney stones (n = 1, 3.3%). The following supplement use was reported: multivitamin (n = 10, 33.3%), vitamin D (n = 9, 30%), calcium (n = 5, 16.7%), and other (n = 8, 2.7%). One subject was lost to follow-up after week 10, but the data were included in the analysis. The remaining 29 of 31 subjects completed the study (retention rate 93.5%). The response rate for symptom surveys was 100%.

Changes in NRS for average pain and ability to return to sport/exercise over the course of the study are shown in Figures 2 and 3. At the end of the study, subjects reported the ability to participate in sport or exercise as follows: 7 (23.3%), same effort and performance with no pain; 1 (3.3%), same effort and performance with pain; 7 (23.3%), same effort but reduced performance; 12 (40.0%), reduced effort and performance; 3 (10.0%), unable to participate in any sport or exercise. Of the

Table 1. Demographics and baseline characteristics

Characteristic	Value
Age, y[a]	35.3 (7.7)
Female, n (%)	21 (70.0)
Race, n (%)	
White	19 (63.3)
Asian	2 (6.7)
Native Hawaiian or Pacific Islander	1 (3.3)
More than 1 race	3 (10.0)
Other	5 (16.7)
Ethnicity, n (%)	
Hispanic	8 (26.7)
Non-Hispanic	22 (73.3)
Location of BSI, n (%)	
Distal tibia	15 (50.0)
Distal fibula	1 (3.3)
Metatarsal	14 (49.7)
Duration of symptoms, weeks[b]	5 [3.125, 7]
NRS average pain score in the past week[a]	5.5 (2.4)
NRS worst pain score in the past week[a]	7.5 (1.7)
Dietary restriction, n (%)	5 (16.7)
Previous BSI, n (%)	9 (30.0)

BSI, bone stress injury; IQR, interquartile range; NRS, numerical rating scale.
[a]Mean (SD).
[b]Median [IQR].

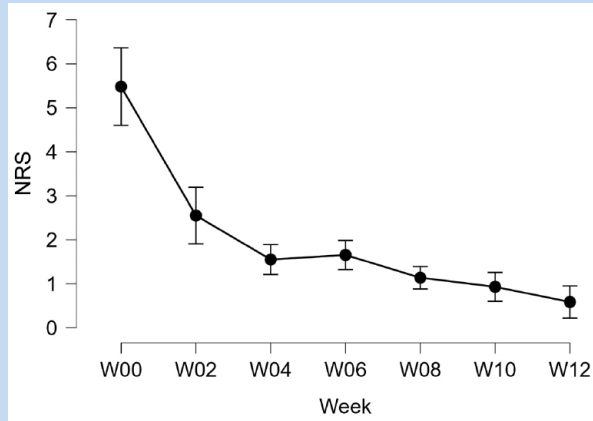

Figure 2. NRS for average pain. Repeated measures ANOVA *P* value <0.01. ANOVA, analysis of variance; NRS, numerical rating scale.

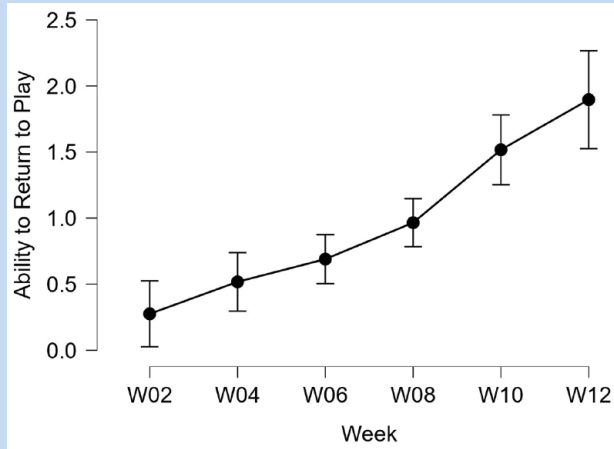

Figure 3. Ability to return to sport/exercise. Return to sport/exercise assessed using a 0 to 4 scale (Methods). Repeated measures ANOVA *P* value <0.01. ANOVA, analysis of variance.

subjects who had not returned to full pain-free participation at week 12, none had achieved this earlier at any point during the study period so the symptoms were attributed to persistence of the original BSI rather than a recurrent injury.

A total of 29 (96.7%) completed the 12-week MRI. Table 2 is a summary of the MRI findings at each time point. Statistically significant decreases in marrow edema and soft tissue edema were observed by MRI between week 0 and week 12 (paired sample *t* test *P* value <0.01 for both). At 12 weeks, cortical fracture remained visible by MRI scan in 8 subjects (26.7%) and bone marrow edema persisted on MRI scan in 16 subjects (53.3%), including moderate or marked edema in 6 subjects (20.1%).

A total of 162/169 (95.9%) of US examinations were completed according to the study protocol; 19 subjects (63.3%) were enrolled within 2 weeks of the MRI scan that diagnosed the BSI and were able to complete an US at week 2. Soft tissue edema, periosteal reaction, and hyperemia were observed in 20 (66.7%), 19 (63.3%), and 13 (43.3%) subjects, respectively. Of the 19 (68.4%) subjects, 13 had at least 1 US finding at week 2 and 25 of 30 (83.3%) had at least 1 US finding at week 4. There was no statistically significant difference in the prevalence of US findings at week 4 based on the affected bone: tibia, 13 of 15 (86.6%), metatarsal 11 of 14 (78.6%), and fibula (1 of 1, 100%) (Chi-squared *P* value 0.76). Callus was observed in only 1 subject, who had a metatarsal stress fracture. Five subjects

Table 2. MRI findings[a]

	Week 0 (n = 30)	Week 12 (n = 29)	*P* value
Marrow edema, mean (SD)	1.3 (0.99)	0.79 (0.86)	<0.01
Soft tissue edema, mean (SD)	1.83 (0.91)	0.41 (0.68)	<0.01
Periosteal reaction, mean (SD)	0.97 (0.72)	1.17 (1.17)	0.26
Cortical fracture, n (%)	13 (43.3)	8 (27.6)	0.27

MRI, magnetic resonance imaging.
[a]Marrow edema, soft tissue edema, and periosteal reaction are reported using a 0 to 3 scale (Methods). Paired sample *t* test was used to assess change in marrow/soft tissue edema and periosteal reaction. McNemar's Chi-square test was used to assess change in cortical fracture.

Table 3. Sonographic findings[a]

	Week 2 (n = 19)	Week 4 (n = 28)	Week 6 (n = 30)	Week 8 (n = 27)	Week 10 (n = 28)	Week 12 (n = 29)	*P* value
Soft tissue edema, mean (SD)	0.95 (0.91)	0.93 (0.77)	0.60 (0.67)	0.44 (0.58)	0.46 (0.58)	0.34 (0.55)	<0.01
Periosteal reaction, mean (SD)	0.63 (0.9)	0.93 (0.86)	0.83 (0.87)	0.74 (0.86)	0.68 (0.77)	0.69 (0.76)	0.23
Hyperemia, mean (SD)	0.63 (0.96)	0.64 (1.1)	0.37 (0.72)	0.26 (0.59)	0.14 (0.45)	0.07 (0.26)	<0.01

ANOVA, analysis of variance.
[a]All parameters are reported using a 0 to 3 scale (Methods). Repeated measures ANOVA was used to assess change in sonographic findings over time.

(16.7%) had no sonographic findings at the site of the BSI at any point during the study, which included 2 subjects with tibia stress fractures and 3 subjects with metatarsal stress reactions. Table 3 shows the mean and SD of soft tissue edema, periosteal reaction, and hyperemia at each time point.

Statistically significant decreases in soft tissue edema, periosteal reaction, and hyperemia were observed by US (repeated measures ANOVA *P* values all <0.01). Persistent sonographic findings were present at WEEK 12: periosteal reaction (16 of 29, 57.1%), soft tissue edema (9 of 29, 31.0%), and hyperemia (2 of 29, 7.1%).

Spearman correlations evaluating the relationship between imaging findings and NRS for average pain and return to sport/exercise at each time point are shown in Tables 4 and 5. Overall correlations were weak, with none exceeding 0.5 and only a few time points when there was a statistically significant correlation. The degree of hyperemia had a statistically significant positive correlation with NRS for average pain at weeks 4 and 6. Similarly, the degree of soft tissue edema by US had a statistically significant positive correlation with NRS for average pain at week 6. The degree of hyperemia also had a statistically significant positive correlation with the ability to return to sport/exercise at week 6. US and MRI assessments were also evaluated for correlation at week 12: periosteal reaction ρ 0.48 (95% CI [0.13, 0.72], *P* value 0.01), and soft tissue edema ρ 0.15 ([-0.23, 0.49], *P* value 0.43). Average pain and return to sport/exercise at week 12 were also analyzed for correlation with baseline MRI, the results of which are found in Online Appendix 4.

Based on the correlation of hyperemia with NRS and return to sport/exercise at week 6, subjects with and without hyperemia at week 6 were compared for NRS average pain and return to sport/exercise in the subsequent weeks (Figure 4). Repeated measures ANOVA between-subjects effects *P* value for NRS for average pain was 0.04 and for the ability to return to sport/exercise was 0.06.

DISCUSSION

In this study, we described the sonographic findings of exercise-related BSIs affecting the distal tibia, distal fibula, and

Table 4. Correlation of sonographic findings with NRS for average pain[a]

	Week 2	Week 4	Week 6	Week 8	Week 10	Week 12
US soft tissue edema, ρ [95% CI]	0.23 [–0.26, 0.61]	0.27 [–0.11, 0.57]	0.38* [0.02, 0.65]	0.29 [–0.1, 0.6]	0.27 [–0.11, 0.58]	0.29 [–0.09, 0.59]
US periosteal reaction, ρ [95% CI]	–0.04 [–0.49, 0.42]	0.26 [–0.12, 0.57]	–0.05 [–0.4, 0.32]	–0.08 [–0.45, 0.32]	–0.36 [–0.64, 0.02]	–0.05 [–0.41, 0.32]
US hyperemia, ρ [95% CI]	0.26 [–0.23, 0.63]	0.45* [0.09, 0.69]	0.42* [0.07, 0.67]	–0.24 [–0.57, 0.16]	0.12 [–0.27, 0.47]	0.35 [–0.03, 0.63]
MRI marrow edema, ρ [95% CI]						0.18 [–0.21, 0.51]
MRI soft tissue edema, ρ [95% CI]						0.20 [–0.18, 0.53]
MRI periosteal reaction, ρ [95% CI]						0.13 [–0.25, 0.47]
MR cortical fracture, ρ [95% CI]						0.39 [0.02, 0.66]

MRI, magnetic resonance imaging; NRS, numerical rating scale; US, ultrasonography.
[a]Spearman correlations (ρ) were used to evaluate the relationship between imaging findings and NRS for average pain at each time point.
*Pvalue <0.05.

metatarsals. Soft tissue edema, periosteal reaction, and hyperemia decreased over the course of 12 weeks. At week 6, increased sonographic soft tissue edema and hyperemia were associated with higher levels of pain (positively correlated), and in those with lower levels of ability to return to sport/exercise (negatively correlated). Increased hyperemia was also similarly associated with higher levels of pain at week 4.

Although the diagnostic value of US for BSIs has been reported previously,[14,22] serial use to monitor BSI healing, correlating with symptoms and return-to-play decision-making, is novel. The previously published data on the use of US to monitor the healing of BSIs are limited to a case report.[2] We demonstrated a statistically significant decrease in soft tissue edema, periosteal reaction, and hyperemia over a 12-week period. This is consistent with the previous case report that described hyperemia and a periosteal reaction on the initial examination, which resolved 7 weeks later.[2] Hyperemia appears to have the most clinical relevance given the positive correlation with pain and negative correlation with the ability to return to sport/exercise. In addition, when differentiating subjects based on the presence or absence of hyperemia at week 6, there was a significant repeated measures ANOVA between-subjects effects for pain. Correlation between sonographic findings and clinical outcomes did not extend beyond week 6, which could be due to the heterogeneity of the study population and variable rates of healing.

The decision regarding the return to sport/exercise after BSI is guided by the location and severity of injury along with clinical symptoms. MRI is the current gold standard for diagnosis of BSI and also offers clinicians valuable information to guide treatment, with MRI scan grade and location of bone injury at initial diagnosis being associated with time out of sport.[13,18] However, the value of repeat MRI to monitor healing can be limited because the resolution of bone edema may lag behind clinically relevant healing and the ability to return to sport/exercise. In our current study, persistent marrow edema by MRI was present in over half of the subjects at 12 weeks and did not correlate with pain or return to sport/exercise. The discrepancy between radiologic and clinical healing is also true for radiography, computed tomography, and bone scintigraphy, thus limiting their value in return-to-sport/exercise decision-making.[3] Therefore, in the absence of reliable, objective measures, the decision of when a patient can safely return to sport/exercise after a BSI is difficult to make with certainty.

Table 5. Correlation of sonographic findings with return to sport/exercise[a]

	Week 2	Week 4	Week 6	Week 8	Week 10	Week 12
US soft tissue edema, ρ [95% CI]	0.04 [–0.42, 0.49]	0.12 [–0.25, 0.46]	–0.15 [–0.48, 0.23]	–0.04 [–0.41, 0.34]	0.18 [–0.21, 0.52]	–0.11 [–0.45, 0.27]
US periosteal reaction, ρ [95% CI]	0.30 [–0.18, 0.66]	–0.22 [–0.54, 0.15]	–0.15 [–0.48, 0.22]	–0.15 [–0.51, 0.24]	0.14 [–0.25, 0.48]	–0.10 [–0.45, 0.28]
US hyperemia, ρ [95% CI]	0.05 [–0.41, 0.49]	–0.10 [–0.45, 0.28]	–0.45* [–0.68, –0.09]	–0.39 [–0.66, 0.01]	–0.26 [–0.57, 0.13]	–0.17 [–0.5, 0.21]
MRI marrow edema, ρ [95% CI]						–0.17 [–0.5, 0.22]
MRI soft tissue edema, ρ [95% CI]						–0.28 [–0.58, 0.11]
MRI periosteal reaction, ρ [95% CI]						0.06 [–0.31, 0.42]
MRI cortical fracture, ρ [95% CI]						–0.22 [–0.54, 0.16]

MRI, magnetic resonance imaging; US, ultrasonography.
[a]Spearman correlations were used to evaluate the relationship between imaging findings and return to sport/exercise at each timepoint. Return to sport/exercise assessed using a 0 to 4 scale (Methods).
*Pvalue <0.05.

This study had several limitations. First, there is currently no standard approach to assessing and quantifying the sonographic findings of BSIs. Given the continuum of BSI severity, we utilized a 4-point Likert scale for soft tissue edema, periosteal reaction, and hyperemia. However, this approach is inherently subjective, creates interobserver variability, and affects reproducibility. Further research is needed to refine the sonographic assessment of BSIs to address these limitations and improve the utility of this modality. Objective sonographic measurements, such as periosteal thickness, could be explored in a more homogenous cohort so that the classifications can be quantified and more reproducible. The sonographic findings and changes over time may be different based on age and location, which affects the generalizability of the results. As for the assessment of MRI scans, previous research of BSI utilizes the Fredericson classification from 1995, which included T1-weighted images.[1,9,18] However, the protocol at our institution is to include fat-suppressed T2 and PD rather than T1 images. The advantage of PD compared with T1-weighted images is the higher resolution and thinner slice thickness, as well as the contrast resolution that allows for better visualization of the fracture line (if present) and periosteal reaction. Whereas MRI scan results were correlated with pain and return to sport/exercise at 12 weeks (Tables 4 and 5), baseline MRI scan results were not correlated with these clinical outcomes as the MRI scans were performed before study enrollment.

This cohort was heterogeneous, with an age range of 23 to 47 years and included injuries of the tibia, fibula, and metatarsal. The sample size was not large enough to control for risk factors for low bone mineral density and use of assistive devices when assessing recovery. In addition, ability to return to sport was assessed using a nonvalidated Likert scale as we were unaware of a relevant validated patient-reported outcome score. Many of the subjects had not returned to full exercise or sport participation without pain by the end of the 12-week study period. Similarly, some subjects still had bone marrow edema and/or a cortical fracture on MRI scan, indicating incomplete healing at 12 weeks. Future studies could collect and evaluate the clinical outcomes and sonographic data beyond 12 weeks. While biweekly examinations may not be logistically feasible in every outpatient setting, US is increasingly available in athletic training rooms, where it could be used for serial monitoring for healing,

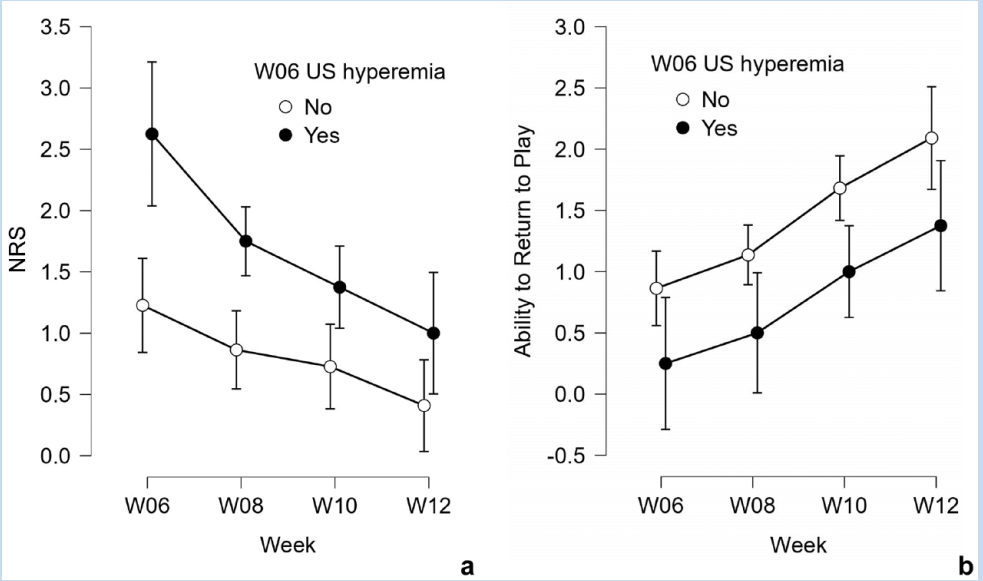

Figure 4. NRS and return to sport/exercise based on hyperemia at week 6. (a) NRS for average pain. (b) Ability to return to sport/exercise. Return to sport/exercise assessed using a 0 to 4 scale (see Methods). Repeated measures ANOVA between-subjects effect *P* value for NRS = 0.04, for return to sport/exercise = 0.06. ANOVA, analysis of variance; NRS, numerical rating scale; US, ultrasonography.

correlation with symptomatology, and aid in return-to-play decisions. Another aspect of assessing BSI healing and readiness to return to sport/exercise that was not included in the current study is the physical examination, eg, tenderness at the site of BSI and pain with hop test. Future studies may benefit from including these clinical outcomes when correlating sonographic findings.

CONCLUSION

Serial US of BSIs can provide objective measures of healing. Hyperemia or soft tissue edema by US may be used in the future to both optimize treatment and more accurately assess readiness for return to sport/exercise after BSIs. Further research is needed to determine the prognostic value of US of BSIs, which may assist sports medicine providers with return-to-sport/exercise decisions.

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